

Agenda

by

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1. AI maturity model (AIMM) :
what and why?
2. AI maturity dimensions
3. How to use the AI maturity
model

When & Why to use the AIMM

When you want to:



Plan your organization's AI strategy, establish AI policy & processes



Set up AI product teams for practicing AI successfully



Check on your organization's AI progress



Provide a map to the new & complex AI territory



Help companies and teams advance on their AI journey by:

- ✓ Identifying where they are
- ✓ Deciding where they want to be

Key Steps for AIMM Assessment



Identify key capabilities of AI Maturity that contributes



Define “Foundational” and “Differentiation” Capabilities – AI differentiation Capabilities are required for reaching higher AI maturity > 30%



Build the AI maturity Index – Measure AI Foundational & Differentiation capability



Construct AI Profile based on foundational and differentiation capabilities



Define & Measure the speed of AI transformation vs. the speed of digital transformation
– Propose Short/Mid term to long term goals

What is in the AI Maturity Model?

Factors that influence AI maturity

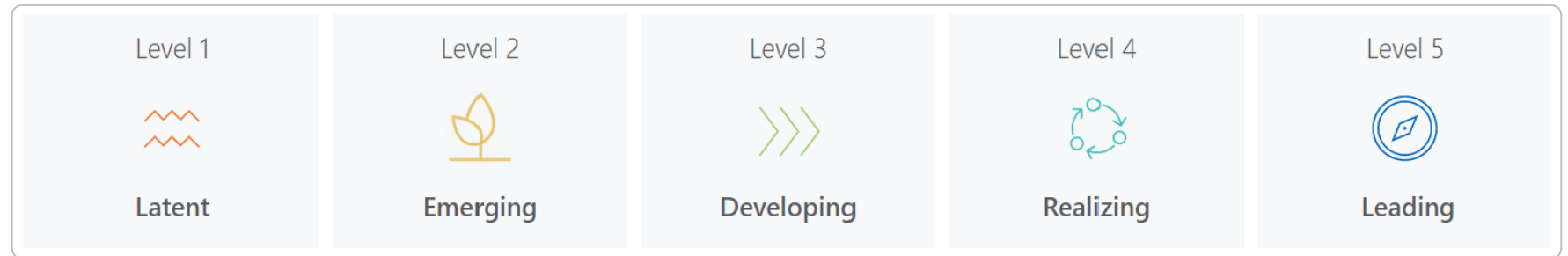
Dimension 1

Dimension 2

Dimension 3

Dimension ...

Maturity levels



In-depth
interviews

AI experts & practitioners
(UST & other companies)

18 months of
research

AIMM Research Methodology based on Becker et al.s and CMMI

The Artificial Intelligence Maturity Model (AIMM) is a framework designed to evaluate and guide organizations in effectively adopting and integrating Artificial Intelligence (AI) technologies

Maturity in the field of AI indicates the degree of development and the strength of the organization's performance measures to mitigate risks that threaten its assets

Development Methodology:

- The creation of the AIMM follows a structured approach based on Becker et al.'s seven-step procedure for developing maturity models:
- **Problem Definition:** Identifying the necessity for a model that assists organizations in assessing and enhancing their AI capabilities.
- **Comparison of Existing Models:** Analyzing current AI maturity models to recognize their limitations and areas lacking theoretical foundation.
- **Development Strategy:** Formulating a plan to construct a new, theoretically sound model, considering the deficiencies identified in existing models.
- **Iterative Maturity Model Development:** Employing an iterative process to design and refine the model, ensuring it meets the practical needs of organizations.

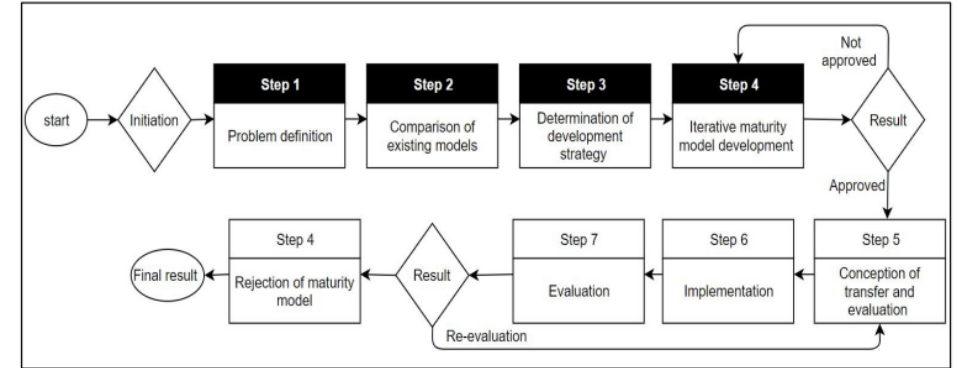
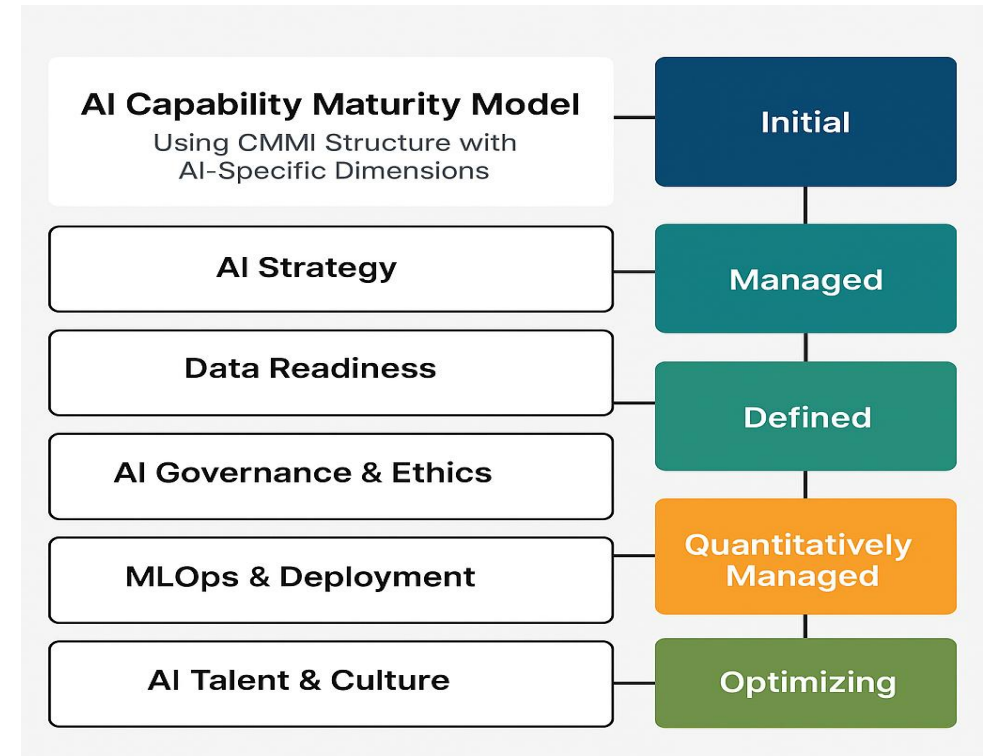


Figure 1. Procedure model for developing a maturity model adopted from (Becker et al. 2009)



AI Maturity Model (AIMM) framework

Table 3. Artificial Intelligence Maturity Model

level	AI functions	Data structure	People	Organisation
Level 1 <i>Initial</i>	Very limited or no AI function exists and has no plans	Regular data structure; no data exists to train AI	Regular IT skills; Organisations lack the skills to evaluate, build and deploy AI solutions	No business case related to AI; existing structure are used informally
Level 2 <i>Assessing</i>	Discovery AI Technology	Integration of current usage of AI into data required to train AI	AI related training; Assessment of existing infrastructure with regard AI	Organisation initial AI strategy; for each AI application, have defined a value proportion
Level 3 <i>Determined</i>	AI project is at an advanced stage; determination of infrastructure needed to further implement AI	Custom AI data are introduced; data standardised are exist	Active management support; resources are provided, AI related employees training	Organisation has standard operating procedures that cover AI scenarios; change management is introduced
Level 4 <i>Managed</i>	AI process are defined throughout the organisations	Appropriate data science exists to make critical business decisions using AI	AI is being fully realised as employees' productivity	There is a well-defined value to support and full top management support
Level 5 <i>Optimise</i>	Full AI infrastructure adoption and standardisation	Proactive data analysis; Data is available in real time	Employees are engaged; centralised leadership	Role, responsibilities and accountability are clearly defined within each AI project; AI culture

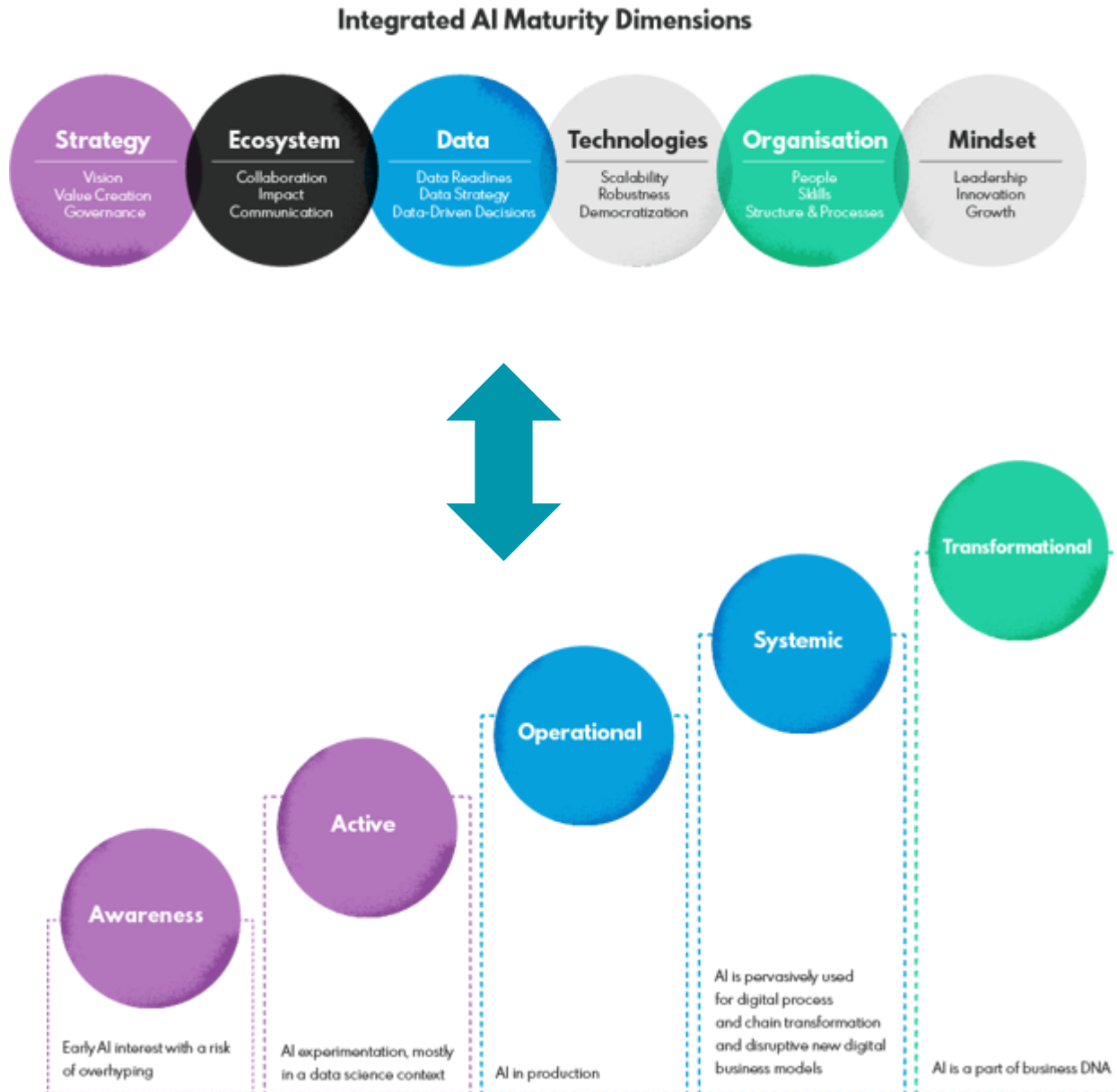
Table 2. Definition of AI dimensions

AI dimension	Definition	Reference
AI functions	AI functionality refers to the tools and technologies that are required to handling AI at scale	Russell et al. (1995)
Data structure	Data refer to containing both the amount and structure of the data to getting AI systems to work by enabling high-velocity capture, discovery or analysis.	Nilsson, (2014)
People	People refers to all those individuals within an organisation to create artificial intelligence technologies.	Rich,(1983); Corea, (2017)
Organisational	Describes business characteristics and resources that might influence AI process such as firm size, managerial structure, decision-making and communication.	Tornatzky and Fleischer (1990)

Start Small, Big Picture & scale fast

Readiness for Redesign

Organizational maturity must be at a certain level to enable proper AI adoption across the whole business.



Key areas to evaluate an organization's AI capabilities:

- Strategy:** Alignment of AI with business goals, leadership commitment, and roadmap.
- Data:** Quality, accessibility, infrastructure, and integration.
- Technology:** Tools, platforms, scalability, and integration with existing systems.
- People & Skills:** Workforce expertise, training, and collaboration.
- Governance:** Ethics, compliance, risk management, and accountability.
- Culture:** Organizational agility, innovation mindset, and stakeholder trust.

A 5-tier model to classify AI maturity:

- 1.**Ad-hoc (Level 1):** No formal strategy; isolated experiments.
- 2.**Emerging (Level 2):** Early pilots, basic tools, and initial governance.
- 3.**Operational (Level 3):** Standardized processes, integrated data/tech, and skilled teams.
- 4.**Strategic (Level 4):** AI-driven decision-making, proactive governance, and cross-functional collaboration.
- 5.**Transformative (Level 5):** AI as a core business driver, ethical innovation, and industry leadership.

GenAI Maturity Check list (Sample)

How to Use This Checklist

1. Workshop Sessions: Collaborate with IT, data, and business teams to score each criterion.

2. Gap Analysis: Highlight "No" and "Partial" responses to identify critical gaps.

3. Roadmap Prioritization: Use the **Priority** column to sequence initiatives (e.g., fix High-priority gaps first).

4. Benchmarking: Reassess quarterly to track progress toward Gen AI maturity.

Category	Criteria	Response (Yes/No/Partial)	Evidence/Comments	Priority
Infrastructure & Cloud				
	1. Scalable cloud infrastructure (AWS, Azure, GCP) with GPU/TPU support for training/inference.		E.g., Use of AWS SageMaker, Azure ML, or Google Vertex AI.	High
	2. MLOps pipelines for model deployment, monitoring, and retraining.		Tools like Kubeflow, MLflow, or Databricks.	High
	3. API integrations to connect Gen AI models with enterprise systems (CRM, ERP).		APIs for Salesforce, SAP, etc.	Medium
	4. Security protocols for data encryption, access control, and compliance (GDPR, HIPAA).		SOC 2 compliance, role-based access control (RBAC).	High
Data Readiness				
	1. High-quality, labeled datasets for training/validating Gen AI models.		Data lakes/warehouses (Snowflake, Redshift) with clean, structured data.	High
	2. Real-time data pipelines for dynamic model inputs (e.g., customer interactions).		Apache Kafka, Spark Streaming, or similar tools.	Medium
	3. Synthetic data generation capabilities to augment training data.		Tools like Gretel.ai, Mostly AI, or custom GANs.	Medium
	4. Metadata management and data lineage tracking.		Collibra, Alation, or custom governance frameworks.	Medium
Process & Automation				
	1. Documented workflows for integrating Gen AI into business processes.		Process maps for customer service, content creation, etc.	High
	2. Existing automation (RPA, workflows) that can integrate with Gen AI outputs.		UiPath, Automation Anywhere, or Zapier integrations.	Medium
	3. Agile change management processes to adapt to AI-driven workflows.		Agile/Scrum teams, feedback loops.	Medium
Digital Ecosystem				
	1. Legacy systems modernized with APIs/microservices for Gen AI compatibility.		Legacy ERP/CRM systems with RESTful APIs.	High
	2. Digital adoption (e.g., customer-facing chatbots, AI-driven analytics tools).		Existing tools like ChatGPT Enterprise, Adobe Firefly.	Medium
Skills & Culture				
	1. AI/ML talent (data scientists, prompt engineers, ethicists) in-house or via partners.		Certifications (AWS ML Specialty, Google Cloud AI Engineer), vendor partnerships.	High
	2. Leadership sponsorship and budget allocated for Gen AI initiatives.		C-suite ownership, dedicated funding.	High
	3. Employee training programs for Gen AI tools (e.g., prompt engineering).		LMS platforms with AI courses (Coursera, Udacity).	Medium
Governance & Ethics				
	1. Framework for ethical AI (bias detection, transparency, fairness).		Tools like IBM AI Fairness 360, Microsoft Fairlearn.	High
	2. Risk mitigation for hallucinations, misinformation, and IP/copyright compliance.		Human-in-the-loop reviews, watermarking AI outputs.	High
	3. Model auditability and version control.		Tools like Weights & Biases, Neptune.ai.	Medium
Use Cases & Strategy				
	1. Gen AI use cases aligned with business goals (e.g., 30% cost reduction in customer service).		Documented ROI metrics, pilot results.	High
	2. Scalable roadmap for enterprise-wide Gen AI adoption.		Phased rollout plans, vendor partnerships (OpenAI, Anthropic).	High

GenAI Model Levels Based on Context Optimization & LLM Optimization

- Generative AI (GenAI) models categorized into levels based on how well they optimize context handling and LLM efficiency.
- Crucial for improving model performance, reducing hallucinations, enhancing personalization, and making responses more accurate and contextually aware.

As **GenAI models** progress from **Level 1 (Basic)** to **Level 5 (AGI-like)**, they improve in **context handling, optimization, memory efficiency, and real-time learning**.

Key challenges includes:

- **Reducing hallucinations** while keeping responses creative.
- **Optimizing token & computational efficiency**.
- **Ensuring bias-free learning** through better alignment techniques.

Level	Context Optimization	LLM Optimization	Key Features	Examples
L1 - Basic	Minimal	Basic	Stateless responses, generic outputs	Simple chatbots, FAQ bots
L2 - Short-Term	Low to Moderate	Intermediate	Session-based memory, context-aware responses	AI customer support, smart search
L3 - Persistent	Moderate to High	Advanced	Long-term memory, fine-tuned LLMs, RAG	AI copilots, virtual assistants
L4 - Adaptive	High	Expert	Real-time adaptation, multi-modal, personalized learning	AI decision-makers, enterprise copilots
L5 - AGI	Extreme	Highly Adaptive	Continuous learning, self-evolving, multi-modal mastery	Artificial General Intelligence (AGI)

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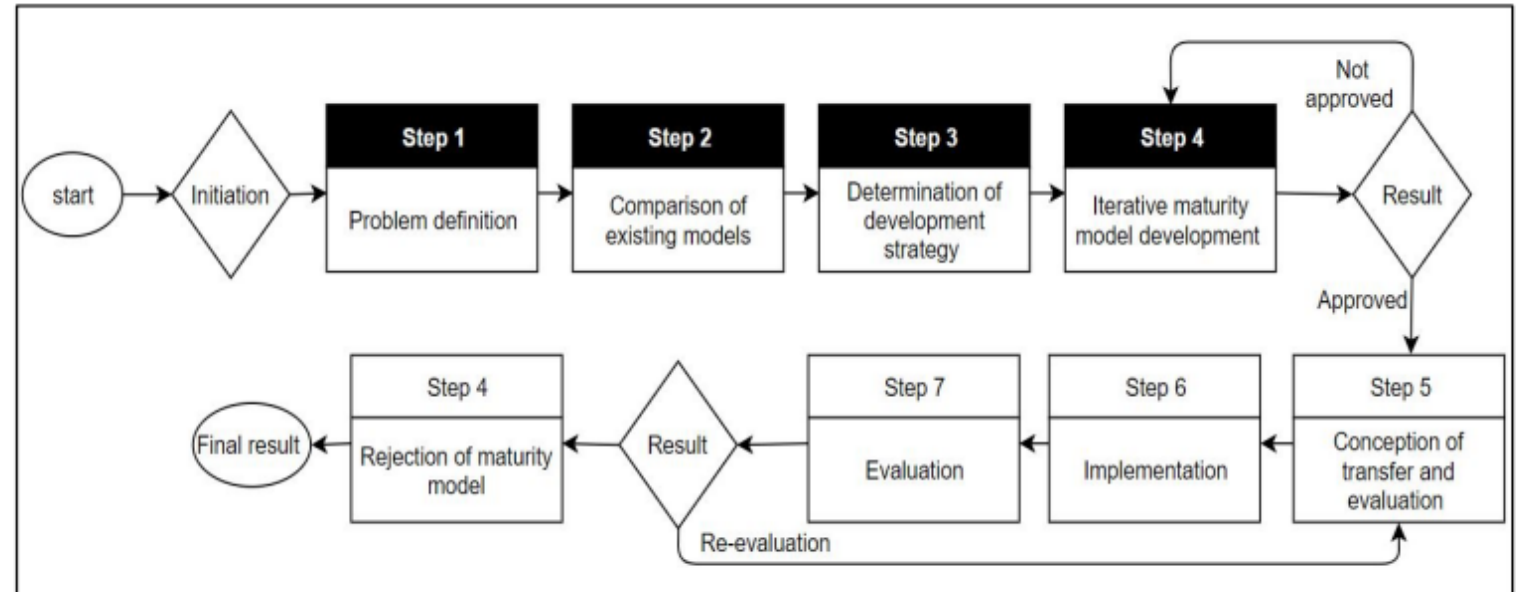


Figure 1. Procedure model for developing a maturity model adopted from (Becker et al. 2009)

Key Capabilities for AI maturity

Strategy and Sponsorship

- 1. Senior Sponsorship: Organizations have an AI strategy that is developed by the Chief Analytics Officer, Chief Data Officer, Chief Digital Officer or an equivalent. The CEO and the Board actively sponsor and share accountability for the strategy and associated AI initiatives.
- 2. AI Strategy: Organizations not only have a core AI strategy aligned to the overall business strategy, but they also dedicate tools and tactics to execute it and continuously track their performance against that strategy.
- 3. Proactive vs. Reactive: Organizations have the resources (such as technology, talent and patents) to proactively define and demonstrate how AI can create value vs. apply AI as a reaction to a need. They're first-movers instead of fast followers in terms of applying AI for business value.
- 4. Readily Available AI and ML tools: Organizations work with an ecosystem of technology partners to access machine learning models and tools to help innovate new products and services.
- 5. Readily Available Developer Networks: Organizations tap into an ecosystem of technology partners to access developer networks that support the development of new products and services.

Data and AI Core

- 6. Build vs. Buy: Organizations develop custom-built AI applications or work with a partner who offers solutions as-a-service, vs. purchase "off-the shelf" AI solutions with little-to-no customization.
- 7. Platform and Technology: Organizations apply the necessary cloud, data and AI infrastructure, software, self-serve capabilities and industry best practices, and they adopt the latest tools available from platform and technology partners.
- 8. Experimentation Data—Change: effectively translating into a higher data and AI maturity. Experimentation data is the use of internal and external data to design new models and generate new insights. to keep data clean and trustworthy, and to support decision making at greater speed and scale.
- 9. Data Management and Governance: Organizations scale their data management and governance practices to increase data quality, trust and ethics across
- 10. Data Management and Governance—Change: Organizations improved their data management and governance

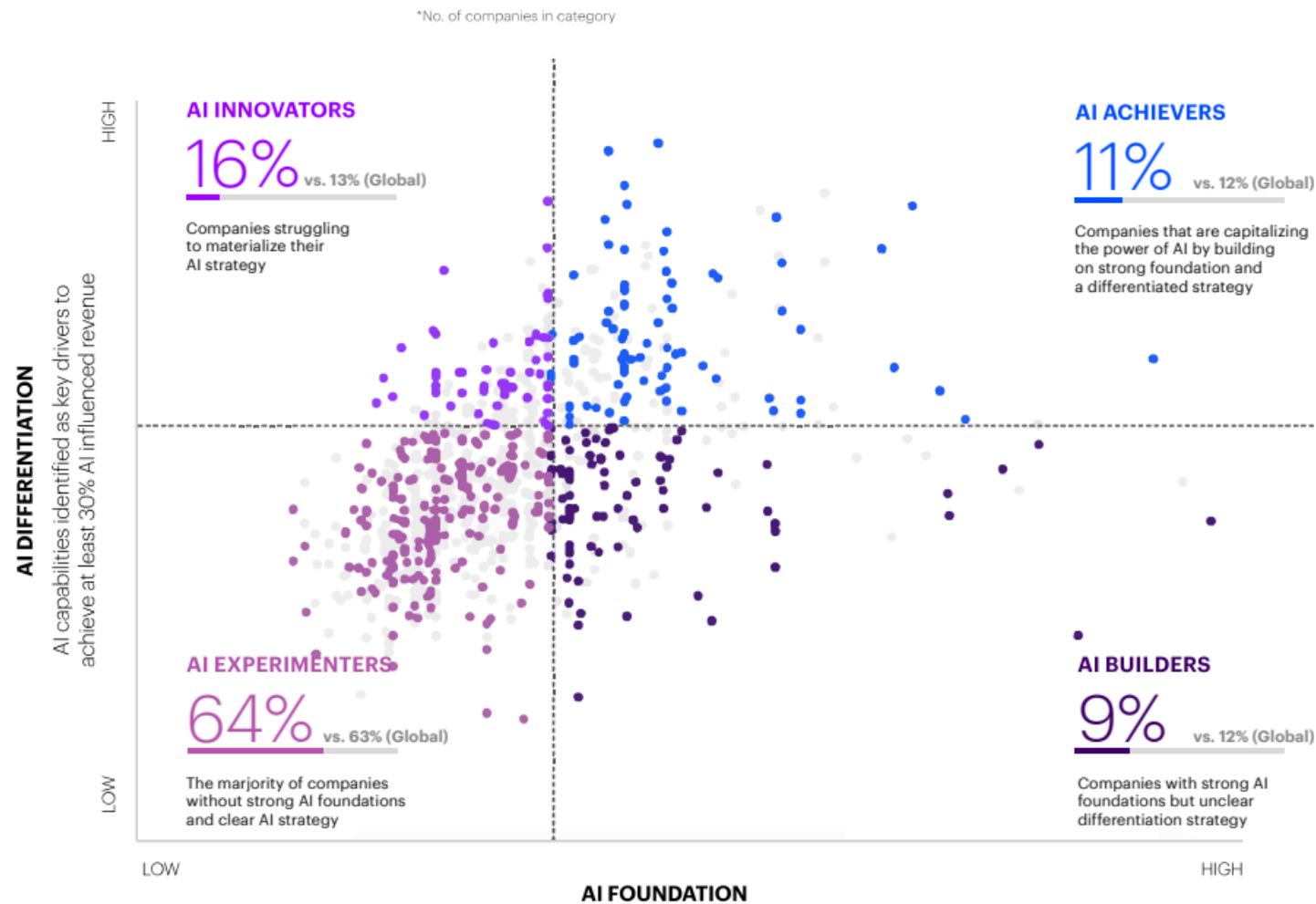
Talent and Culture

Employee Competency, Innovation Culture & AI talent Strategy

Responsible AI

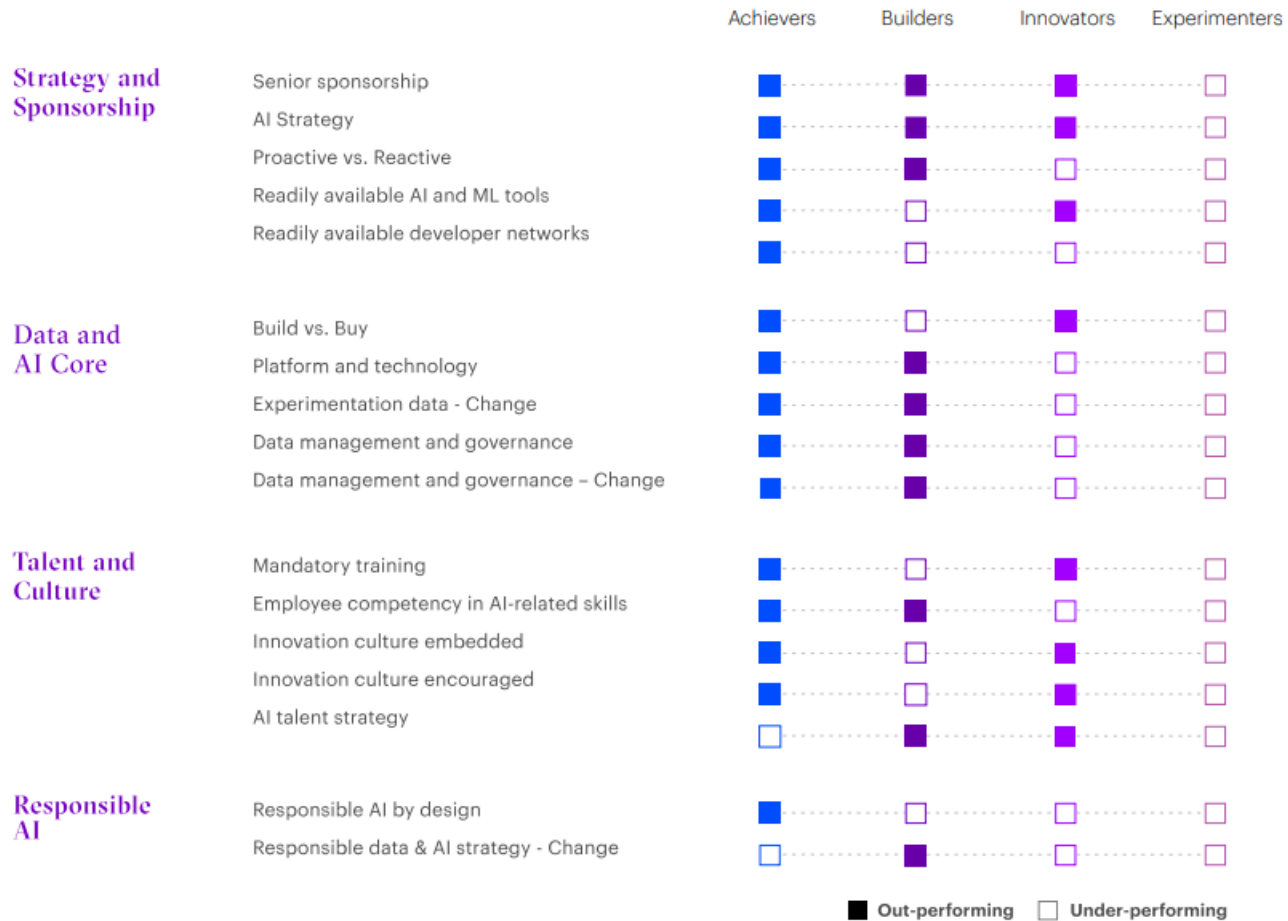
Approach to address changing regulatory requirements, mitigate risks, and support sustainable, trustworthy AI

Figure 3: Only 11% of organizations are AI Achievers



Source: Accenture Research analysis based on a sample of 1200 global companies of which 422 are from Europe (including Germany, France, Spain, Italy and UK)

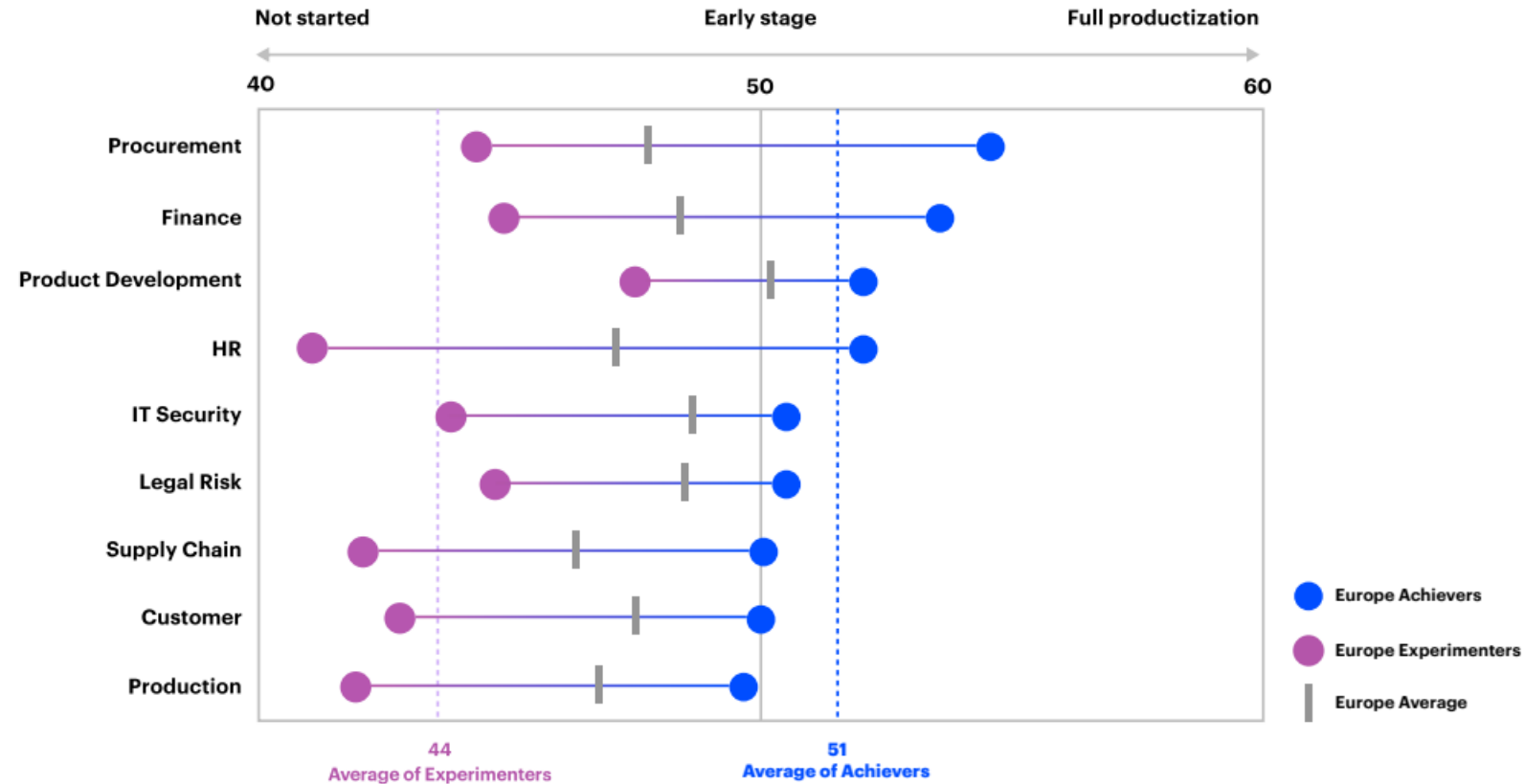
Figure 5: AI Achievers outperform in nearly all capabilities



Source: Accenture Research

Note: Each cube represents one of the 17 key capabilities. The cube is highlighted when the AI profile is outperforming against peers (higher than the average across all companies in terms of % of companies reaching the mature level).

Figure 6: Achievers excel at turning AI pilots into production



Source: Accenture Research

Note: Score 0-100, ranging from 0 = AI use case not started, 50 = AI use in early stage, 100 = having AI programs in place for full productization. The chart shows the difference in terms of average score for AI use cases of different functions, between Achievers and other firms. Those differences are statistically significant after controlling for industry, geography, and company size; see Appendix for more details.

Custom AI model development

Major Steps

1. Problem Definition & Requirement Analysis

1. **Objectives:** Clearly define the business or technical problem and expected outcomes.
2. **Stakeholders:** Identify the key stakeholders and decision-makers.
3. **Success Criteria:** Establish performance metrics and evaluation benchmarks.

2. Data Acquisition & Preprocessing

1. **Data Collection:** Gather data from relevant internal and external sources.
2. **Data Cleaning:** Remove noise, handle missing values, and ensure quality.
3. **Data Transformation:** Normalize, scale, and structure data for model consumption.
4. **Exploratory Data Analysis (EDA):** Understand patterns, correlations, and insights.

3. Model Design & Selection

1. **Algorithm Choice:** Evaluate and select machine learning algorithms (e.g., regression, classification, deep learning).
2. **Feature Engineering:** Identify and construct relevant features.
3. **Architecture Design:** Define the model architecture, layers, and hyperparameters.

4. Model Training & Validation

1. **Training:** Train the model using the prepared dataset.
2. **Validation:** Use cross-validation, hold-out sets, or A/B testing to assess performance.
3. **Hyperparameter Tuning:** Optimize parameters to enhance accuracy.
4. **Iteration:** Refine the model based on validation feedback.

5. Deployment & Integration

1. **Deployment Strategy:** Choose between cloud, on-premise, or hybrid deployment.
2. **Integration:** Incorporate the model into existing applications or workflows.
3. **API/Interface:** Develop endpoints for model interaction.
4. **Scalability:** Ensure the solution can handle production load.

6. Monitoring & Maintenance

1. **Performance Monitoring:** Set up dashboards and alerts to track real-time performance.
2. **Feedback Loop:** Collect user and system feedback for continuous improvement.
3. **Model Retraining:** Schedule updates as new data becomes available or when performance degrades.
4. **Compliance & Security:** Ensure ongoing data privacy and adherence to regulations.

Prerequisites

- **Clear Requirements:** Well-defined objectives and use cases.
- **Data Availability:** Access to quality, relevant data along with necessary permissions.
- **Technical Resources:** Adequate computational resources, hardware, and software tools.
- **Expertise:** Collaboration among data scientists, engineers, domain experts, and IT staff.
- **Infrastructure:** Established data pipelines, storage, and processing capabilities.
- **Governance:** Data security, privacy policies, and regulatory compliance measures.

Deliverables

• Documentation:

- Requirements specification document.
- Data collection and preprocessing reports.
- EDA and feature engineering insights.
- Model design and architecture documentation.
- Training logs and validation reports.

• Artifacts:

- Cleaned and processed datasets.
- Trained model (code, weights, configurations).
- APIs or deployment packages.
- Monitoring dashboards and alert configurations.

• Operational Plans:

- Deployment and integration strategies.
- Maintenance and retraining schedules.
- User guides and technical manuals.

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